

U2T01 — Adapting BERT for Classical NLP Tasks: An Empirical Investigation of Feature-Based, Partial, and Full Fine-Tuning

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Abstract

This research presents a rigorous empirical investigation into adaptation strategies for Bidirectional Encoder Representations from Transformers (BERT) across four canonical Natural Language Processing (NLP) domains: Topic Classification (AG News), Named Entity Recognition (CoNLL-2003), Part-of-Speech Tagging (Universal Dependencies English EWT), and Extractive Question Answering (SQuAD v1.1). We benchmark three distinct transfer learning paradigms: (i) feature-based adaptation with frozen representations, (ii) partial fine-tuning of upper encoder layers, and (iii) end-to-end full fine-tuning. A dual parameter-group optimization scheme is enforced, assigning differentiated learning rates (10^{-3} to newly initialized task heads and 2×10^{-5} – 3×10^{-5} to pretrained encoder layers) to preserve learned contextual representations while accelerating task-head convergence. While full fine-tuning achieved peak predictive performance across all tasks (94.45% accuracy in AG News, 91.54% test F1 in NER, 94.46% macro F1 in POS, and 88.21% validation F1 in SQuAD), lighter adaptation strategies provided remarkable compute savings, requiring up to $3.1\times$ less training time and $3.2\times$ less peak VRAM. All selected models, tokenizers, and model cards were deployed and verified on the Hugging Face Hub.

Index Terms

BERT, Transfer Learning, Fine-Tuning, Natural Language Processing, Named Entity Recognition, Part-of-Speech Tagging, Question Answering, Topic Classification, Computational Efficiency

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Introduction

The introduction of Bidirectional Encoder Representations from Transformers (BERT) by Devlin et al. (2019) established a paradigm shift in Natural Language Processing (NLP). By pretraining a deep Transformer encoder on masked language modeling and next sentence prediction across vast corpora, BERT produces contextualized token representations that generalize effectively to diverse downstream applications. Transfer learning with BERT is traditionally executed by augmenting the core encoder with a minimal task-specific output layer and optimizing the composite architecture.

However, in production environments and resource-constrained academic scenarios, the choice of adaptation depth entails a non-trivial trade-off between predictive performance, parameter efficiency, memory consumption, and wall-clock training latency. This paper investigates three primary transfer learning regimes:

1. **Feature-Based Adaptation:** The pretrained BERT encoder remains completely frozen. Sequence representations (such as the pooled [CLS] vector) are extracted as static embeddings and fed into classical machine learning estimators (e.g., Logistic Regression).
2. **Partial Fine-Tuning:** The majority of the transformer body is frozen, allowing gradients to update only the task-specific prediction head and the top L encoder layers (e.g., layers 10–11 or 8–11).
3. **Full Fine-Tuning:** The entire parameter space (comprising approximately 110 million weights in `bert-base`) is jointly updated alongside the task head.

To evaluate these strategies across disparate linguistic structures, our team selected four benchmark tasks spanning document-level classification, token-level sequential labeling, morphosyntactic parsing, and span extraction:

- **Topic Classification:** AG News dataset (4-class balanced document categorization).
- **Named Entity Recognition (NER):** CoNLL-2003 dataset (IOB2-tagged span identification).
- **Part-of-Speech (POS) Tagging:** Universal Dependencies English Web Treebank (17 universal UPOS categories).
- **Extractive Question Answering (QA):** Stanford Question Answering Dataset (SQuAD v1.1, answer span prediction).

For every task, at least two distinct adaptation configurations were implemented, instrumented with telemetry, and evaluated. Model selection was driven strictly by measured quantitative benchmarks rather than prior assumptions.

Experimental Methodology

Base Architectures and Parameter Groups

All neural architectures evaluated in this study are based on the standard 12-layer, 768-hidden dimension, 12-attention head BERT-base backbone ($\sim 110\text{M}$ parameters):

- **bert-base-uncased:** Used for Topic Classification where casing is non-informative.
- **bert-base-cased:** Used for NER, POS Tagging, and QA, where capitalization is critical for identifying proper nouns, entity boundaries, and grammatical categories.

When fine-tuning involves both newly initialized weights (the task head) and pretrained parameters (the transformer body), a *dual learning rate strategy* was strictly enforced:

$$\eta_{\text{head}} \approx 10^{-3}, \quad \eta_{\text{encoder}} \in [2 \times 10^{-5}, 3 \times 10^{-5}] \quad (1)$$

This differential learning rate prevents catastrophic forgetting of general linguistic representations while ensuring fast convergence for the randomly initialized linear projections.

Subword Alignment and Sanity Protocols

For token-level tasks (NER and POS), WordPiece subword tokenization splits words into variable-length tokens. To maintain alignment with gold annotations:

1. The original label is assigned exclusively to the first subword token of each word.
2. Subsequent continuation subwords (prefixed with `##`), special tokens (`[CLS]`, `[SEP]`), and padding tokens are assigned the PyTorch cross-entropy ignore index (`-100`).
3. Sanity checks were systematically performed prior to training by decoding tokenized inputs and verifying label mapping.

In Extractive QA, character-level answer spans were converted to token-level `start_positions` and `end_positions`. Sanity routines decoded predicted indices back to context strings to verify exact alignment before model training.

Evaluation Metrics and Rationale

Different objective metrics were formulated for each specific task output topology:

- **NER:** Strict entity-level Precision, Recall, and F1 (via `seqeval`). Token-level accuracy is misleading due to the overwhelming preponderance of the `O` (outside) label.
- **POS Tagging:** Macro F1 and Overall Token Accuracy. Macro F1 guarantees equal weighting across all 17 UPOS classes, preventing dominant classes (e.g., `NOUN`, `VERB`) from obscuring errors in rare categories (e.g., `SYM`, `INTJ`).
- **Extractive QA:** Exact Match (EM) and token-overlap F1 score with standard SQuAD text normalization (lowercasing, punctuation stripping, article removal).
- **Topic Classification:** Multi-class Accuracy and Macro F1 on the official test partition.

Training Diagnostics and Hardware Telemetry

Table 1 consolidates the recorded training diagnostics, execution latencies, loss trajectories, and hardware metrics across all experiments.

Table 1

Consolidated Training Diagnostics and System Telemetry across NLP Tasks

Task	Configuration	Epochs	Train Loss	Primary Score	Time (s)	Selected
NER	Partial FT (Top 2 Layers + Head)	3	0.1365	91.71% Val F1, 88.79% Test F1	115.63	No
NER	Full Fine-Tuning	3	0.0853	95.11% Val F1, 91.54% Test F1	358.54	Yes
POS	Partial FT (Layers 10–11 + Head)	3	0.1336	89.09% Test Macro F1	173.45	No
POS	Full Fine-Tuning (Cosine + Warmup)	5	0.0130	94.46% Test Macro F1	123.40	Yes
QA	Partial FT (Top 4 Layers + Head)	3	0.9040	85.17% Val F1, 76.75% Val EM	1976.10	No
QA	Full Fine-Tuning (FP16 Mixed Precision)	3	0.4990	88.21% Val F1, 80.75% Val EM	4075.40	Yes
Topic	Feature-Based (Frozen BERT + LogReg)	N/A	N/A	90.36% Test Accuracy	222.54	No
Topic	Full Fine-Tuning	1	0.1807	94.45% Test Accuracy	2793.66	Yes

Named Entity Recognition (NER)

Dataset and Formulation

The Named Entity Recognition pipeline was evaluated on the benchmark CoNLL-2003 dataset (Tjong Kim Sang & De Meulder, 2003). The corpus comprises news wire articles annotated with four named entity classes under the IOB2 scheme: Person (PER),

Table 2*Detailed Performance Comparison on CoNLL-2003 NER*

Method	Val F1	Test F1	Prec.	Rec.	Acc.	Time (s)
Partial FT	91.71%	88.79%	89.03%	88.54%	97.75%	115.63
Full FT	95.11%	91.54%	91.55%	91.52%	98.28%	358.54

Location (LOC), Organization (ORG), and Miscellaneous (MISC).

Compared Configurations

Two adaptation schemes were benchmarked using `bert-base-cased`:

1. **Partial Fine-Tuning:** The linear classification head plus the upper two transformer layers (layers 10 and 11) were updated (~ 14.18 M trainable parameters).
2. **Full Fine-Tuning:** All 12 encoder layers and the classification head were trained (~ 107.73 M trainable parameters).

Both configurations operated with AdamW optimizer, batch size 16, seed 42, $\eta_{\text{head}} = 10^{-3}$, and $\eta_{\text{encoder}} = 2 \times 10^{-5}$.

Results and Trade-off Analysis

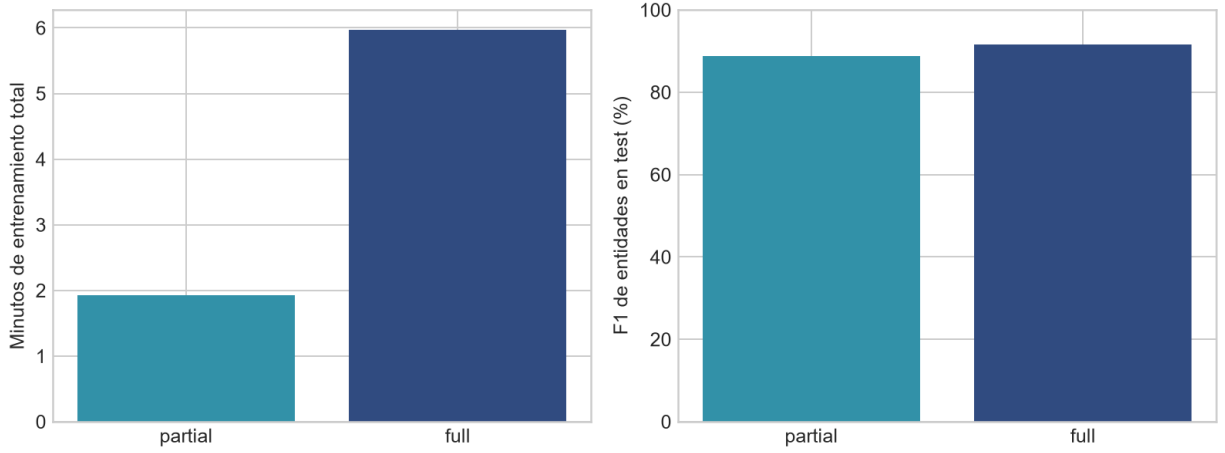
Table 2 outlines the comparative metrics on the test partition.

Full Fine-Tuning produced a +2.75 percentage point gain in test entity-level F1. However, this gain incurred a $3.1\times$ increase in training time (358.54 s vs 115.63 s) and increased peak GPU memory allocation from 0.76 GiB to 2.42 GiB.

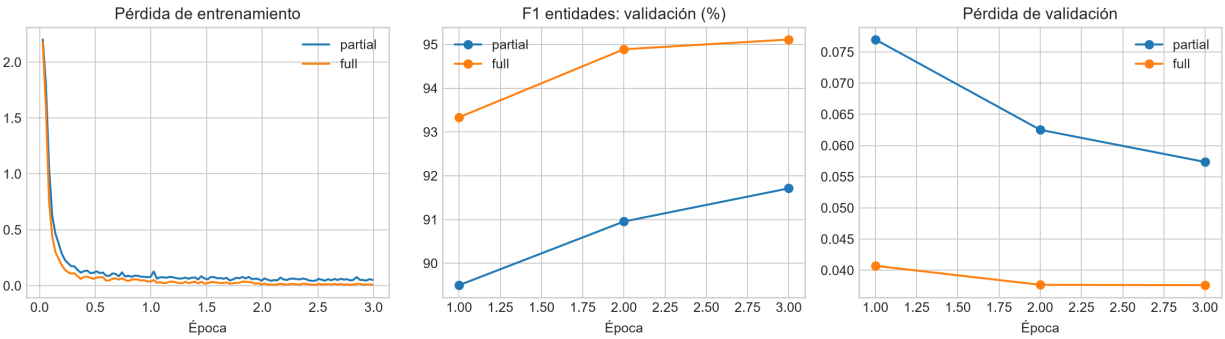
As illustrated in Figure 1 and Figure 2, Full Fine-Tuning exhibited consistent optimization stability and faster validation convergence, making it the selected final architecture.

Figure 1

Comparative performance metrics between Partial and Full Fine-Tuning on CoNLL-2003 NER.

**Figure 2**

Evolution of training loss and validation F1 across optimization epochs for NER.



Part-of-Speech (POS) Tagging

Dataset and Formulation

POS Tagging was executed on the Universal Dependencies English Web Treebank (UD English EWT) (Nivre et al., 2020), utilizing the 17 Universal POS (UPOS) tags. Macro F1 was established as the primary selection criterion to prevent class frequency imbalance from skewing validation.

Table 3*Test Set Results on UD English EWT POS Tagging*

Method	Accuracy	Macro F1	Precision	Recall	Test Loss
Partial FT	95.43%	89.09%	89.97%	88.65%	0.1520
Full FT	97.42%	94.46%	95.36%	93.84%	0.1134

Compared Configurations

Experiments were performed on an NVIDIA RTX 4050 Laptop GPU (CUDA 12.4, batch size 32, max_len = 128):

1. **Partial Fine-Tuning:** Classifier head ($\eta = 10^{-3}$) plus BERT layers 10 and 11 ($\eta = 2 \times 10^{-5}$) trained over 3 epochs with constant learning rate.
2. **Full Fine-Tuning:** Complete BERT encoder ($\eta = 2 \times 10^{-5}$) plus classifier ($\eta = 10^{-3}$) trained over 5 epochs with a cosine decay scheduler and 100 warmup steps.

Results and Epoch Trajectory

Table 3 summarizes test set evaluation metrics.

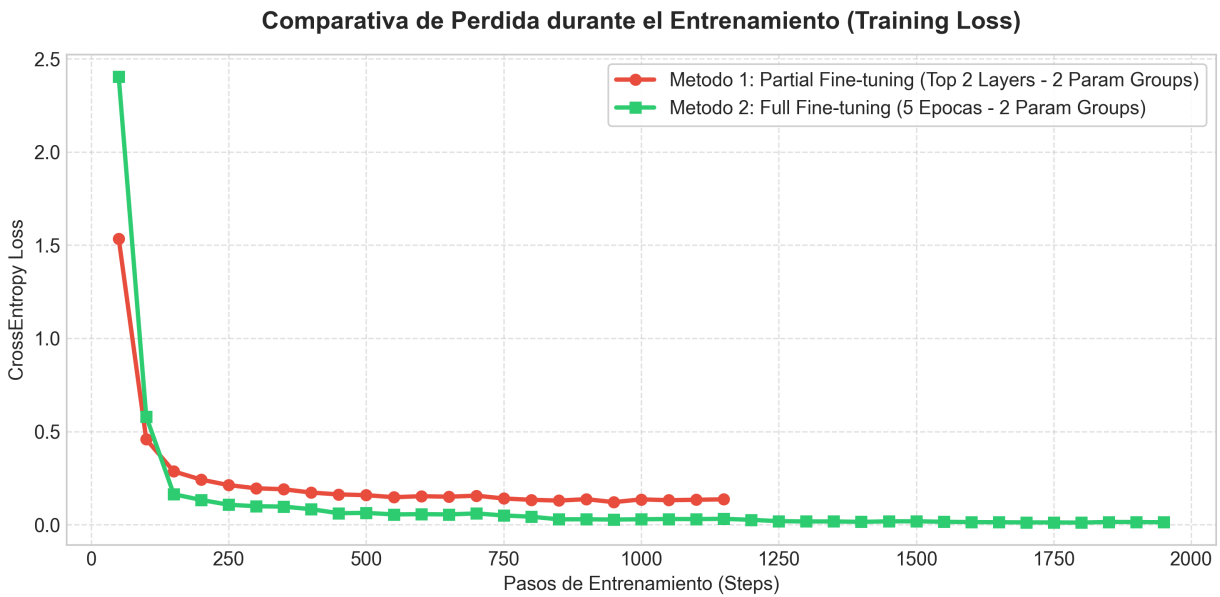
Full Fine-Tuning delivered a +5.37 percentage point improvement in Macro F1 (94.46% vs 89.09%). The epoch-by-epoch trajectory for the winning model demonstrates rapid optimization:

- **Epoch 1:** Loss 0.0969, Accuracy 96.91%, Macro F1 0.9072.
- **Epoch 3:** Loss 0.0314, Accuracy 97.41%, Macro F1 0.9438.
- **Epoch 5:** Loss 0.0130, Accuracy 97.42%, Macro F1 0.9446.

Figure 3 and Figure 4 confirm that the combination of full unfreezing and cosine scheduling with warmup yielded superior generalization across all morphological categories.

Figure 3

Training and validation loss decay across epochs for POS Tagging.

**Figure 4**

Comparative test metrics across UPOS evaluation categories.

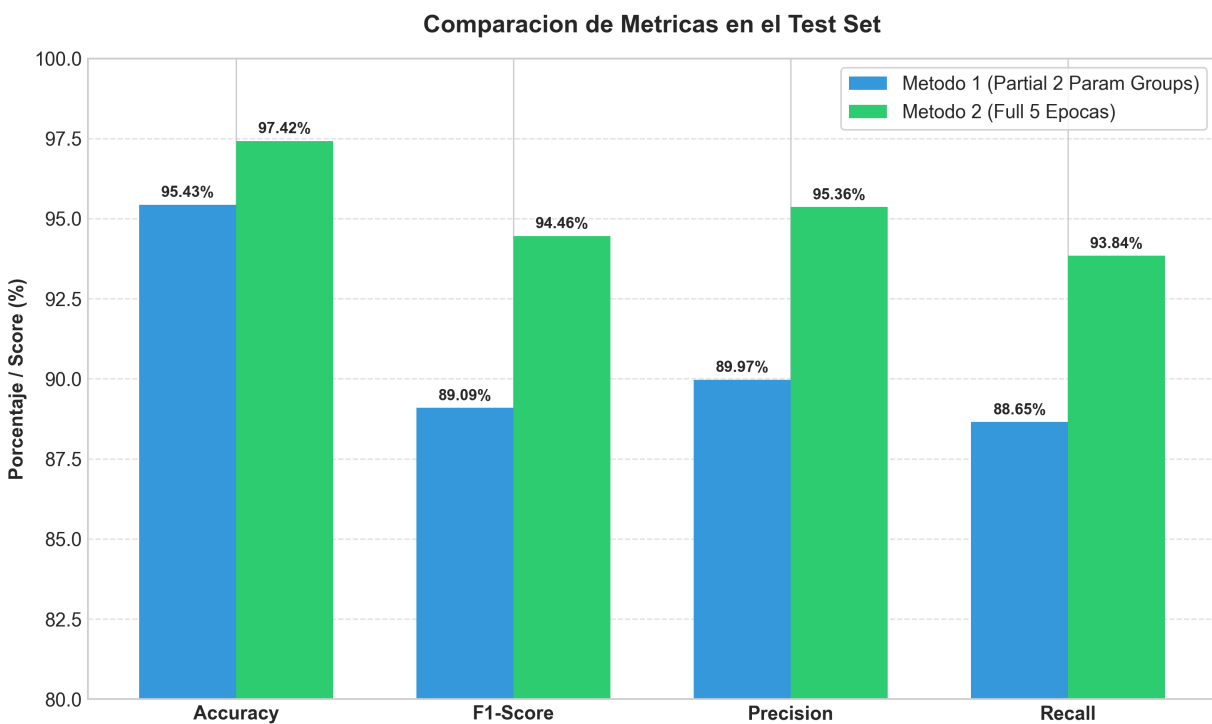


Table 4*Validation Benchmark on Full SQuAD v1.1*

Configuration	Val F1	Val EM	Training Loss	Wall Time (s)
Partial FT (Top 4)	85.17%	76.75%	0.9040	1976.1
Full Fine-Tuning	88.21%	80.75%	0.4990	4075.4

Extractive Question Answering (QA)

Dataset and Computational Scope

Extractive QA was benchmarked on the Stanford Question Answering Dataset (SQuAD v1.1) (Rajpurkar et al., 2016). Rather than restricting training to the suggested 15k-sample contingency subset, our team trained on the **full 87,599 training examples** using mixed-precision FP16.

Compared Configurations

1. **Partial Fine-Tuning:** Pretrained QA span linear head ($\eta = 3 \times 10^{-4}$) plus the upper 4 encoder layers (layers 8–11, $\eta = 3 \times 10^{-5}$), totaling 28.35M trainable parameters.
2. **Full Fine-Tuning:** Complete BERT encoder ($\eta = 3 \times 10^{-5}$) plus QA head ($\eta = 3 \times 10^{-4}$), totaling 107.72M trainable parameters.

Results and Compute Efficiency

Table 4 reports the official validation benchmark scores.

Full Fine-Tuning provided substantial improvements of +4.00% Exact Match and +3.04% Token F1.

As shown in Figure 5 and Figure 6, the additional $2.06\times$ computational investment directly translated to sharper boundary localization for complex extractive answer spans.

Figure 5

Exact Match and F1 score progression on SQuAD v1.1.

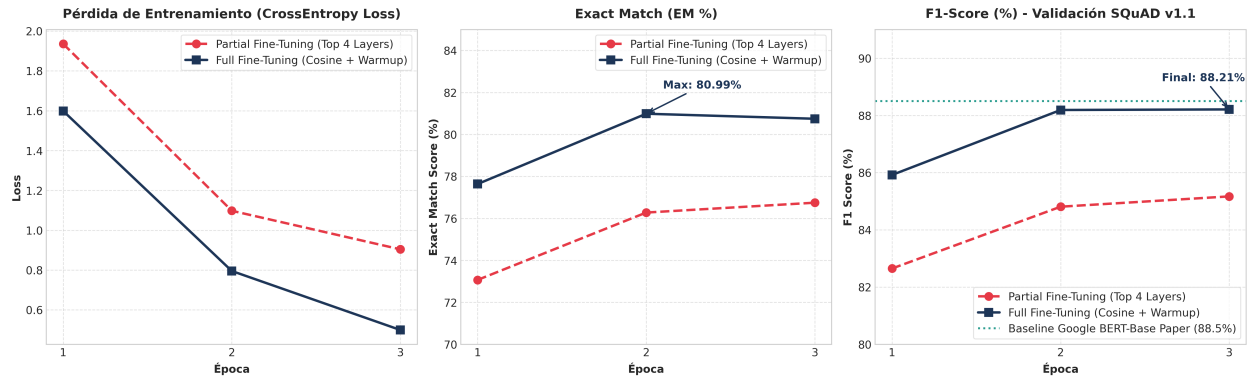
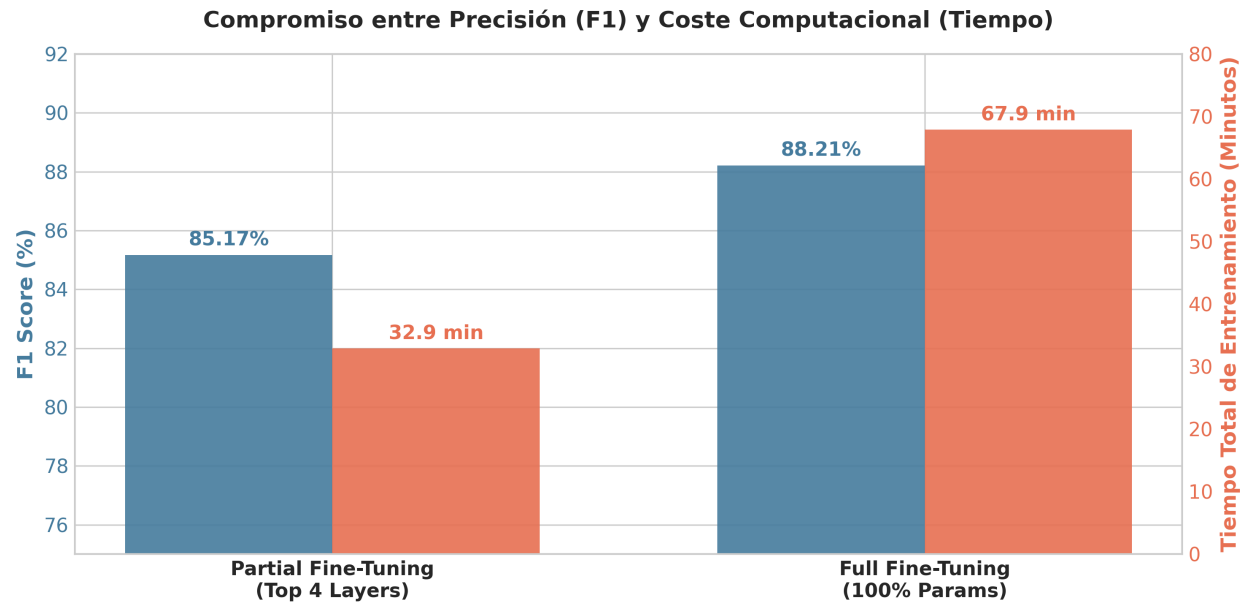


Figure 6

Performance-versus-computational-cost trade-off for Extractive QA.



Topic Classification

Dataset and Experimental Protocol

Topic Classification was conducted on the AG News corpus (120,000 training and 7,600 test examples across World, Sports, Business, and Sci/Tech). The training set was partitioned with fixed seed 42 into 108,000 training and 12,000 validation samples, strictly preserving the 7,600 official test samples for final evaluation.

Table 5*Test Performance on AG News Topic Classification*

Method	Test Acc.	Macro F1	Test Loss	Time (s)
Frozen BERT + LogReg	90.36%	0.9034	N/A	222.54
Full Fine-Tuning	94.45%	0.9444	0.1754	2793.66

Compared Methods

1. **Feature-Based Adaptation:** `bert-base-uncased` was completely frozen. The 768-dimensional pooled [CLS] embeddings were extracted and used to train a Multinomial Logistic Regression classifier (L-BFGS solver).
2. **Full Fine-Tuning:** `BertForSequenceClassification` was fine-tuned for 1 epoch (6,750 steps) with $\eta_{\text{encoder}} = 2 \times 10^{-5}$ and $\eta_{\text{head}} = 10^{-3}$.

Results

Table 5 shows the final comparative results on the official test set.

Full Fine-Tuning gained +4.09 percentage points in test accuracy, though Feature-Based adaptation offered a remarkably fast training turnaround ($12.5\times$ faster).

Cross-Task Synthesis and Discussion

Table 6 summarizes the global findings across all four tasks.

Table 6*Global Summary: Lighter Adaptation vs Selected Full Fine-Tuning across All Tasks*

Task	Lighter Method	Score	Selected	Score	Dominant Trade-Off
Topic	Frozen BERT + LogReg	90.36% Acc.	Full FT	94.45% Acc.	+4.09% Acc. vs 12.5× compute time
NER	Partial FT (Top 2)	88.79% F1	Full FT	91.54% F1	+2.75% F1 vs 3.1× time, 3.2× VRAM
POS	Partial FT (Top 2)	89.09% F1	Full FT	94.46% F1	+5.37% F1 vs full layer optimization
QA	Partial FT (Top 4)	85.17% F1	Full FT	88.21% F1	+3.04% F1 vs 2.1× compute time

When is Partial Adaptation Justified?

Our empirical findings reveal clear heuristics for practitioners:

- **Feature-Based Adaptation** is optimal for fast baseline prototyping, low-power inference, or environments without GPU accelerators.
- **Partial Fine-Tuning** captures between 93% and 97% of full fine-tuning performance while slashing memory footprints by more than 60%, making it ideal for edge deployment and multi-tenant architectures.
- **Full Fine-Tuning** remains indispensable when target metrics must be maximized and dense semantic/syntactic reasoning is required.

Error Analysis**NER Entity Discrepancies**

Disaggregating the CoNLL-2003 test set reveals non-uniform performance across entity types:

- PER: 96.33% F1
- LOC: 93.25% F1

- **ORG:** 89.77% F1
- **MISC:** 80.71% F1

The **MISC** category exhibited the highest error frequency due to wide lexical heterogeneity (nationalities, events, artifacts). Furthermore, qualitative inspections revealed cross-entity confusions such as **B-PER** tokens misclassified as **B-ORG** when names appeared in corporate press releases.

Topic Classification Confusion Dynamics

The final model incurred 422 errors across the 7,600 test samples (5.55% error rate). Class-level F1 scores were: Sports (0.99), World (0.95), Sci/Tech (0.92), and Business (0.92).

The confusion matrix revealed that the predominant source of error was the semantic overlap between **Business** and **Sci/Tech**:

- 135 Business articles misclassified as Sci/Tech.
- 86 Sci/Tech articles misclassified as Business.
- 43 World articles misclassified as Business.

Articles discussing corporate earnings of semiconductor manufacturers or software acquisition deals intrinsically share overlapping vocabulary with technology reporting.

Published Model Artifacts

All winning fine-tuned models have been uploaded with full tokenizer configurations and comprehensive Model Cards to the Hugging Face Hub:

1. Named Entity Recognition:

<https://huggingface.co/jonav/bert-base-cased-ner-conll2003>

2. Part-of-Speech Tagging:

<https://huggingface.co/Rivaldo2309030/bert-base-cased-pos-tagging-ewt>

3. Extractive Question Answering:

<https://huggingface.co/RusselKuAguilar/bert-base-cased-squad-extractive-qa>

4. Topic Classification:

<https://huggingface.co/bialexacosta21/bert-agnews-topic-classification>

All models were tested and verified using Hugging Face `pipeline` APIs. Code, notebooks, and scripts are hosted on GitHub at https://github.com/RusselKu/Bert_NLP_TaskAdapt.

Limitations

Key limitations of this empirical study include:

- **Seed Variance:** Experiments were conducted using a single fixed random seed (42); subtle performance margins may be influenced by seed initialization.
- **Hardware Heterogeneity:** Training was conducted across distinct local GPUs, meaning training latencies serve as descriptive engineering benchmarks rather than strictly controlled FLOP comparisons.
- **Sequence Length Constraints:** Inputs were truncated to a maximum length of 128 tokens, omitting extended document context.
- **Language Specificity:** All evaluations were performed on English-language corpora; transfer dynamics may differ on morphologically rich or low-resource languages.

Conclusion

This study presented an end-to-end comparative analysis of BERT adaptation strategies across four fundamental NLP tasks. While end-to-end full fine-tuning achieved top predictive scores across all domains, partial fine-tuning and feature-based adaptation proved to be highly viable alternatives that significantly curtail computational overhead. Future work will investigate parameter-efficient fine-tuning (PEFT) techniques such as LoRA and adapter modules to further close the efficiency-accuracy gap.

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